

GRA-Swarm: Scaling Structural Reasoning and Collective Novelty in Swarms of AI Agents

Oleg Bit — GRA Ecosystem / qqewq

2026

Abstract

This paper presents the **GRA-Swarm** architecture — a distributed system of AI agents in which collective reasoning is built not on simple aggregation of answers, but on generating alternative hypotheses, detecting contradictions, performing structural revisions, and subsequent verification.

The central thesis of this work is that scaling collective intelligence is determined not by the absolute number of agents, but by the number of *truly independent and verified directions of structural search*. We formalize the concepts of effective number of agents, structural diversity, saturation point (N_{sat}), and Effective Cognitive Throughput (ECT), moving the discussion of “exponential intelligence growth” from intuitive speculation into a rigorous mathematical framework.

1. Introduction: The Scaling Problem in Multi-Agent Systems

The intuitive assumption that increasing the number of agents linearly or exponentially improves task performance encounters fundamental limitations. If agents use identical models, data, and strategies, **correlation of outputs** arises, leading to duplicated work and growing communication costs.

The **GRA (Gödel-Reductio ad Absurdum)** methodology treats contradiction not as an error, but as an object of computation. The basic GRA cycle is:

$$\text{State} \xrightarrow{\text{detect}} \text{Contradiction} \xrightarrow{\text{critique}} \text{Structural Revision} \xrightarrow{\text{verify}} \text{New State}$$

In a swarm, many such cycles execute in parallel. However, the key question is not how many agents work on a task, but how many *truly independent useful solution paths* exist within a given swarm.

2. Mathematical Model of GRA-Swarm

To rigorously describe swarm dynamics, we introduce a formal apparatus.

2.1. Hypothesis Space and Structural Revisions

Let S be the current state space. Each agent $i \in \{1, \dots, N\}$ generates a set of structural revisions $R_i = \{r_{i,1}, \dots, r_{i,m}\}$.

The potential combinatorial space of all possible revisions grows exponentially:

$$|R_{\text{total}}| \sim m^N$$

However, potential space is not equivalent to the space of *useful* solutions. A significant portion of combinations is redundant or technically infeasible.

2.2. Effective Number of Agents and Correlation

Two agents are useful as independent researchers only if their hypotheses are orthogonal. We introduce a similarity measure (correlation) between hypotheses of agents i and j as $\rho_{ij} = \text{sim}(h_i, h_j)$. Let the average correlation in the swarm be denoted as ρ .

The effective number of independent agents N_{eff} (analogous to design effect in sampling statistics) is computed as:

$$N_{\text{eff}} = \frac{N}{1 + (N - 1)\rho}$$

Implications of the formula:

1. As $\rho \rightarrow 0$: $N_{\text{eff}} \approx N$ (ideal scaling).
2. As $\rho \rightarrow 1$: $N_{\text{eff}} \rightarrow 1$ (the swarm degenerates into a single agent with N copies).

Scaling the number of agents and scaling their independence are not the same thing.

2.3. Filtering for Novelty and Usefulness

Pure generation of combinations does not constitute scientific novelty. We introduce three sequential filters for a hypothesis h :

1. **Correctness (C):** h is non-contradictory. $C(h) \in \{0, 1\}$.
2. **Novelty ($\mathcal{N}$):** h differs from known alternatives. $\mathcal{N}(h) = 1 - \max_{h' \in H_{\text{known}}} \text{sim}(h, h') > \theta_{\text{nov}}$.
3. **Usefulness ($\mathcal{U}$):** h improves the objective functional J . $\mathcal{U}(h) = \Delta J(h) > \theta_{\text{use}}$.

The set of useful hypotheses is:

$$H_{\text{useful}} = \{h \in R_{\text{total}} \mid C(h) \wedge \mathcal{N}(h) \wedge \mathcal{U}(h)\}$$

Growth of $|R_{\text{total}}|$ may be exponential, but the dynamics of $|H_{\text{useful}}|$ follow an entirely different law due to filtering.

2.4. Saturation Point N_{sat} and Net Performance

Any real architecture eventually encounters saturation. Let the measurable quality of the swarm be $Q(N)$.

The marginal gain from adding agents is: $\Delta Q(N) = Q(N) - Q(N - 1)$.

The saturation point N_{sat} is defined operationally:

$$N_{\text{sat}} = \min\{N \mid \Delta Q(N) < \epsilon \text{ for } k \text{ consecutive steps}\}$$

where ϵ is the significance threshold and k is the stabilization window. N_{sat} is not a universal constant; it depends on topology, budget, and task complexity.

Since swarm operation incurs costs, we introduce **Net Performance**:

$$\text{Net}(N) = Q(N) - \lambda \cdot \text{Cost}(N)$$

where $\text{Cost}(N) = C_{\text{comp}} + C_{\text{comm}} + C_{\text{ver}} + C_{\text{coord}}$, and λ is a resource cost weighting coefficient.

2.5. Effective Cognitive Throughput (ECT)

To evaluate practical efficiency, we introduce the **ECT (Effective Cognitive Throughput)** metric:

$$\text{ECT}(N) = \frac{|H_{\text{useful}}(N)|}{\text{Cost}(N)}$$

The optimal swarm size N_{opt} is found as:

$$N_{\text{opt}} = \arg \max_N \text{ECT}(N)$$

This allows distinguishing three regimes: efficiency growth ($N < N_{\text{opt}}$), saturation, and degradation ($N \gg N_{\text{opt}}$).

3. Architecture and Topology

3.1. Communication Constraints and Hierarchy

In a flat swarm (all-to-all), the number of communications grows quadratically: $O(N^2)$.

GRA-Swarm employs hierarchical topologies (trees, sparse graphs, clusters), reducing complexity to $O(N \log N)$ or $O(N)$.

Hierarchy performs three functions:

1. **Communication compression:** Lower-level agents explore local optima, middle levels aggregate patterns, upper levels work with meta-hypotheses.
2. **Knowledge abstraction:** Transition from local reasoning to collective and then to meta-reasoning.

3.2. Components of the GRA Ecosystem

- **GRA-Multiverse:** Explores not one but multiple alternative task models $\{M_1, \dots, M_k\}$. Each branch evolves independently, after which GRA evaluates their compatibility.
- **GRA-Hammer:** A structural shock operator $\mathcal{H} : S \rightarrow S'$. While ordinary optimization changes the *state* within space S , Hammer changes the *search space itself* S (e.g., changes data representation or introduces a new class of variables).
- **Validator / Rollback:** A protection mechanism against accepting destructive changes.

$$\text{State}_{\text{new}} = \text{Apply}(\text{State}_{\text{old}}, \Delta S)$$

If $J(\text{State}_{\text{new}}) < J(\text{State}_{\text{old}})$ by independent criteria, a **Rollback** is executed.

- **GRA-Conveyor:** Forms an evolutionary history of search, preserving successful structural transitions and reasons for rejection for use in future iterations.

4. Experimental Research Program

To verify the theory, a series of controlled experiments is proposed.

4.1. Testing the Exponential Regime

Hypothesis: within a certain range of N , quality grows exponentially:

$$Q(N) \approx Q_0 e^{\alpha N} \implies \ln Q(N) \approx \ln Q_0 + \alpha N$$

To avoid overfitting, the exponential growth model must be compared with alternatives (linear, power-law, logarithmic, saturating $Q_{\text{max}}(1 - e^{-\beta N})$) using AIC/BIC criteria on held-out samples.

4.2. Diversity and Topology Control

Experiments must measure N_{eff} , $|H_{\text{useful}}|$, and N_{sat} for groups with low, medium, and high diversity, as well as for various topologies (all-to-all, ring, tree, hierarchical).

Expected result: excessive diversity leads to loss of coordination; there exists an optimal level of diversity that maximizes ECT.

4.3. Parallel and Sequential Tasks

Scalability must be evaluated as a function of task structure T :

$$\text{Scalability}(T) = \frac{Q_{\text{swarm}}(T)}{Q_{\text{single}}(T) \cdot \text{Cost}_{\text{overhead}}}$$

For easily decomposable tasks, scaling is effective; for strictly sequential tasks, swarm growth may degrade performance due to synchronization delays.

5. Applied Aspects and Fundamental Limits

5.1. Scientific and Mathematical Discovery

In automated scientific research, GRA-Swarm distributes roles: hypothesis generation, counterexample search, numerical modeling, novelty checking, and formal verification.

In mathematical tasks, if one strategy (symbolic derivation) reaches a dead end, **GRA-Hammer** changes the object representation (switching to numerical search or alternative topology), exploring not only the solution space but also the space of *representation methods*.

5.2. Three Types of Stability

For autonomous systems, it is critical to distinguish:

1. **Optimization stability:** residing in a local minimum of J .
2. **Dynamic stability:** return to an attractor after perturbation.
3. **Robust stability:** maintaining functionality under significant changes in inputs or environment.

5.3. Fundamental Limits

It is necessary to recognize four classes of constraints that cannot be overcome by simply adding agents:

1. **Architectural:** correlation, redundancy, verification cost.
2. **Computational:** time, memory, bus bandwidth.
3. **Physical:** energy consumption, heat dissipation, computational density.
4. **Logical:** Gödel's incompleteness theorems, algorithmic undecidability (Turing limits). GRA cannot compute what is fundamentally uncomputable.

6. Conclusion

GRA-Swarm does not claim that “more agents = more intelligence,” nor does it postulate a universal exponential dependence of quality on N .

The main scientific result of this work is the following formulation:

GRA-Swarm creates a space of collective structural revisions, potentially growing combinatorially. The real useful effect of this space is determined by the effective number of independent agents, structural diversity, verification quality, and communication costs. Under favorable conditions, there exists a segment of accelerated (including exponential) growth of useful novelty, which is eventually replaced by architectural saturation (N_{sat}).

GRA-Swarm transforms from a hypothesis about “exponential intelligence” into a rigorous research program studying how the architecture of structural revisions turns a multitude of AI agents into a scalable system of collective search, capable of restructuring its own solution space.